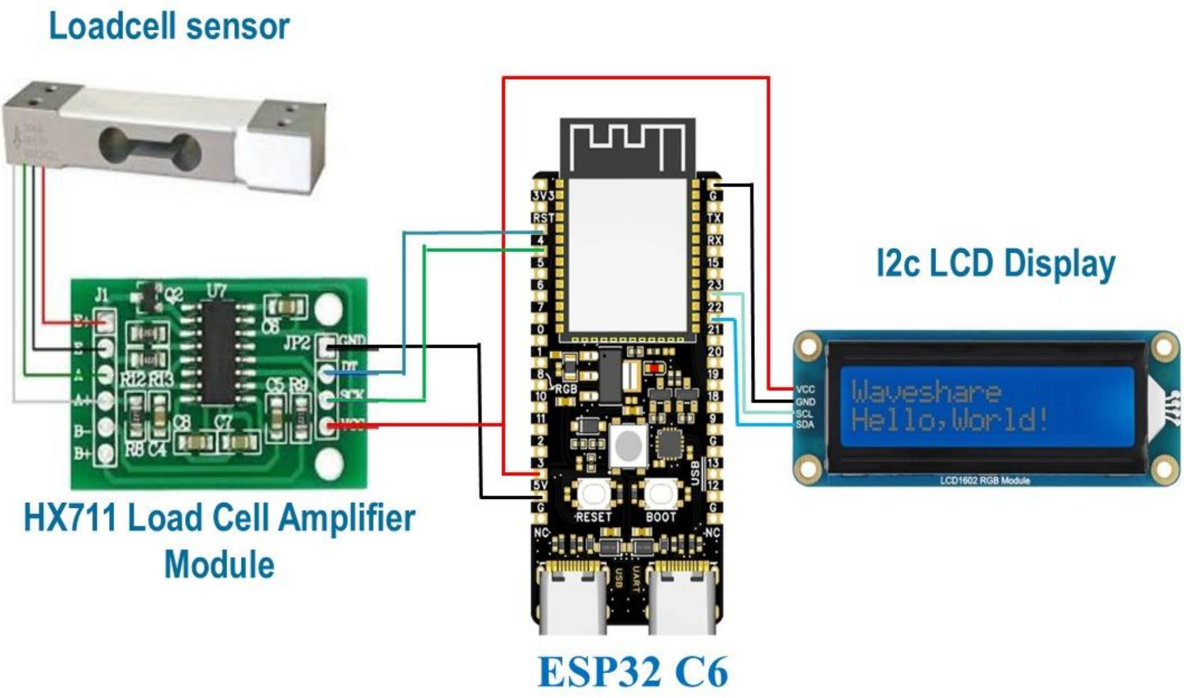
Gram-mar



Aim:

The aim of this project is to design and develop a digital weighing system using an ESP32- microcontroller, HX711 load cell amplifier, and a load cell sensor to accurately measure weight in grams. The measured weight is processed by the ESP32 and displayed in real time on a 16×2 LCD display, while also being transmitted to the Serial Monitor for debugging and calibration.

Components Required:

S.No	Component Name	Quantity
1.	ESP32C6 Dev Module	1
2.	HX711 Load Cell Amplifier Module	1
3.	Load Cell Sensor	1
4.	16×2 LCD Display with I2C Module	1
5.	Breadboard	1
6.	Jumper Wires	As Required

Procedure – Detailed Wiring Connections

1. Load Cell Connections

   1. Connect the Red wire of the load cell to the E+ terminal of the HX711 module.
   2. Connect the Black wire of the load cell to the E− terminal of the HX711 module.
   3. Connect the White wire of the load cell to the A− terminal of the HX711 module.
   4. Connect the Green wire of the load cell to the A+ terminal of the HX711 module.

2. HX711 Load Cell Amplifier Module Connections

   1. Connect the VCC pin of the HX711 module to the 5V pin of the ESP32-C6.
   2. Connect the GND pin of the HX711 module to the GND pin of the ESP32-C6.
   3. Connect the DT (DOUT) pin of the HX711 module to GPIO 4 of the ESP32-C6.
   4. Connect the SCK (CLK) pin of the HX711 module to GPIO 5 of the ESP32-C6.

3. 16×2 I2C LCD Display Connections

   1. Connect the VCC pin of the LCD module to the 5V pin of the ESP32-C6.
   2. Connect the GND pin of the LCD module to the GND pin of the ESP32-C6.
   3. Connect the SDA pin of the LCD module to GPIO 21 of the ESP32-C6.
   4. Connect the SCL pin of the LCD module to GPIO 22 of the ESP32-C6.

4. ESP32-C6 Power Connections

    1. Power the ESP32-C6 using a USB cable connected to a computer or a 5V USB adapter.
    2. Ensure that all modules share a common ground (GND) with the ESP32-C6.
    3. Verify all wiring connections before powering on the circuit.

5. System Verification

    1. Connect the ESP32-C6 to the computer using a USB cable.
    2. Upload the program using the Arduino IDE.
    3. Open the Serial Monitor and set the baud rate to 115200 bps.
    4. The LCD will display the initialization message.
    5. After initialization, the LCD and Serial Monitor will display the measured weight in grams.
    6. Place a known weight on the load cell to verify the accuracy of the measurement.
    7. If necessary, adjust the calibration factor in the program until the displayed value matches the actual weight.

Full code:

```
#include "HX711.h"
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

#define DOUT 4
#define CLK  5

HX711 scale;

// LCD Address (Try 0x3F if 0x27 doesn't work)
LiquidCrystal_I2C lcd(0x27, 16, 2);
```

```
float calibration_factor = -15.26;

void setup() {

  Serial.begin(115200);

  scale.begin(DOUT, CLK);
  scale.set_scale(calibration_factor);
  scale.tare();

  // LCD
  lcd.init();
  lcd.backlight();

  lcd.setCursor(0,0);
  lcd.print("Weight Scale");
  lcd.setCursor(0,1);
  lcd.print("Initializing");

  delay(2000);
  lcd.clear();
}

void loop() {

  long sum = 0;

  for (int i = 0; i < 20; i++) {
    sum += scale.get_units();
    delay(10);
  }

  float weight = sum / 20.0;

  if (abs(weight) < 2)
    weight = 0;

  // Serial Output
  Serial.print("Weight: ");
  Serial.print(weight, 1);
  Serial.println(" g");

  // LCD Output
```

```
  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("Weight:");

  lcd.setCursor(0,1);
  lcd.print(weight, 1);
  lcd.print(" g");

  delay(500);
}
```